

Mario Suarez Ayala

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Robotics • Embedded Systems • Electronics • Mechanics • Renders

PROJECT DOCUMENTATION

Project	Project Vulcan
Version	1.0.0
Date	May 5, 2026
Author	Mario Suarez Ayala
Repository	github.com/mario-suarez-ayala/[repo]
Type	Mechanical
Status	Stable



Project Overview

Project Vulcan is a mechanical engineering project focused on the design and fabrication of a heavy-duty industrial tool cart. The system was developed using carbon steel (14 gauge) to ensure structural strength, durability, and mobility in workshop environments. The objective was to create a functional and resistant tool storage solution capable of safely supporting heavy loads while maintaining stability and maneuverability.

Technical Specifications

Fill in the table with the key system parameters. Add or remove rows according to the project type.

Parameter	Value / Description	Unit
Microcontroller / SBC	Not applicable	—
Operating Voltage	Not applicable	V
Maximum Current	Not applicable	A
Clock Frequency	Not applicable	MHz
Communication Protocol	Not applicable	—
Dimensions (L × W × H)	Approx. 800 × 500 × 900	mm
Weight	Approx. 35–45	kg
Degrees of Freedom (DOF)	Not applicable	—
Software / Firmware	Not applicable	—

3D Design Tool	SolidWorks	—
Render Engine	Blender Cycles	—

System Architecture

Mechanical Subsystem

The structure was built using 14-gauge carbon steel, featuring a reinforced frame with angular supports and multiple levels for tool organization. The design includes a wide base and low center of gravity for stability, along with industrial wheels for mobility.

Manufacturing processes included:

- Cutting (angle grinder)
- Bending (manual press)
- Welding (SMAW with 6013 and 7018 electrodes)
- Surface finishing and painting
-

Electronic Subsystem

- Not applicable.

Software / Firmware Subsystem

- Not applicable.

Render / Visualization

- The model was designed in SolidWorks and later rendered in Blender. The render setup included industrial-style materials and lighting to simulate a realistic workshop environment.

Bill of Materials (BOM)

Component	Specification	Qty	Reference
Carbon Steel Sheets	14 gauge (~1.9 mm)	—	—
Industrial Wheels	25 kg capacity each	6	—
Welding Electrodes	E6013 / E7018	—	—
Steel Angles & Supports	Structural reinforcement	—	—
Paint	Anti-corrosive red finish	—	—

Installation & Usage

- Fabricate the structure following CAD design
- Assemble and weld all structural components
- Install wheels and verify alignment
- Apply surface finishing and paint

- Load tools ensuring proper weight distribution



Results & Testing

The final system achieved a stable and functional structure capable of supporting up to 120 kg of distributed load with a safety margin. The cart demonstrated good maneuverability, structural integrity, and durability under typical workshop conditions.



Contributing

This project was developed as a team effort, with contributions in design, manufacturing, structural analysis, and finishing processes.



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Contact

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